Supplementary Material for

"The Price of Stale State: When Does Network Awareness Pay in LLM Inference Serving?"

This supplement contains the three tables that the main text refers to as Tables S1–S3. Table S1 characterizes the three workload sources. Table S2 counts, per source and per discounting rule, how often the rule beats both the full-trust and the network-blind extremes. Table S3 reports the one-factor-at-a-time sensitivity of the per-channel rule. The tables are provided here, rather than in the main text, only to keep the main manuscript within the target length; they are not additional experiments.

S1. Load characterization

Table S1: Load characterization per source (mean over segments). "CV(1s)" is the coefficient of variation of per-second arrival counts; "AC(1s)" the lag-1 autocorrelation of that series; "rate chg." the mean relative change of the 5-second windowed arrival rate across a 5-second shift; "Fano" the variance-to-mean ratio of per-second counts. A stationary Poisson process would show CV ≈ 0, AC ≈ 0 and Fano ≈ 1.

Source	Workload	CV(1s)	AC(1s)	Rate chg.	Fano
Synthetic	mixed	0.60	0.85	0.30	7.0
	data-heavy	0.65	0.73	0.27	3.7
Azure	mixed	0.23	0.17	0.14	1.4
	data-heavy	0.30	0.28	0.17	1.6
BurstGPT	mixed	1.30	0.73	0.43	302
	data-heavy	2.11	0.55	0.99	939

S2. Verdict counts for the discounting rules

S3. One-factor-at-a-time sensitivity of the per-channel rule

Table S2: Verdict counts over 24 conditions per source (6 staleness configurations × 2 workloads × 2 network conditions). "Better than both" counts conditions where the rule beats *both* full trust and no trust. No rule dominates.

Source	Rule	beats both	loses both	beats full	loses full
Synthetic	per-channel	5	1	5	18
	term-split	4	3	4	19
	uniform	5	5	5	18
	hard	5	1	5	17
Azure	per-channel	1	4	1	22
	term-split	1	5	1	22
	uniform	2	4	4	19
	hard	4	8	7	15
BurstGPT	per-channel	8	2	10	13
	term-split	6	2	8	15
	uniform	10	4	10	13
	hard	7	3	8	14

Table S3: One-factor-at-a-time ablation of the per-channel rule (mean Δp95 vs. the unmodified rule, with 95% CI over 12 conditions; "flips" counts conditions where the sign of "rule vs. full trust" changes).

Variant	Synthetic	Azure	BurstGPT	flips (max)
$\tau_q : 4 \to 2\,\mathrm{s}$	$+6.3 \pm 8.4$	$+2.6 \pm 2.9$	$+1.9 \pm 3.4$	1/12
$\tau_q : 4 \to 8\,\mathrm{s}$	-3.2 ± 3.9	-1.0 ± 1.4	-0.9 ± 1.4	1/12
$\tau_n : 60 \to 20\,\mathrm{s}$	$+1.3 \pm 1.7$	$+0.4 \pm 0.6$	$+1.0 \pm 1.5$	0/12
$\tau_n : 60 \to 120\,\mathrm{s}$	-1.2 ± 1.8	-0.7 ± 0.8	-0.8 ± 1.1	2/12
floor: $0.7 \to 0.5$	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0/12
floor: $0.7 \to 0.9$	-1.6 ± 1.9	-0.8 ± 0.9	-0.9 ± 1.2	2/12